

Program : Diploma in Chemical Engineering	
Course Code : 6072C	Course Title: Energy Conservation & Audit (CH)
Semester : 6	Credits: 4
Course Category: Open Elective	
Periods per week: 4 (L: 3 T: 1 P: 0)	Periods per semester: 60

Course Objectives:

- To enable the students to identify the needs for energy management in minimizing energy wastage

Course Pre-requisites:

Topic	Course code	Course name	Semester
Environmental Science		Environmental Science	2

Course Outcomes :

On completion of the course student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Summarize the need for energy management.	12	Understanding
CO2	Summarize energy management strategies	18	Understanding
CO3	Identify the energy conservation plan and best financial investment.	18	Understanding
CO4	Summarize the energy auditing methods, Sankey diagram and total energy system.	12	Understanding

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1					3		
CO2					3		
CO3					3		
CO4					3		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcome	Description	Duration (Hours)	Cognitive Level
CO1	Summarize the need for energy management.		
M1.01	Recall the available energy resources and its classification	3	Understanding.
M1.02	Summarize the conventional and non-conventional energy resources.	3	Understanding.
M1.03	Enumerate the need for energy reforms and energy planning	3	Understanding.
M1.04	Outline energy conservation act 2001	3	Understanding
Contents: Introduction to energy scenario- world energy resources and their classification- Conventional and non-conventional resources with examples -Indian scenario for renewable and non-renewable resources Need for energy reforms-energy reforms in India Energy conservation act 2001			
CO2	Summarize energy management strategies		
M2.01	Summarize the methodology for energy analysis for production process	2	Understanding
M2.02	Recall the laws of thermodynamics and energy balance	3	Understanding
M2.03	Enumerate the uses of energy measurement by online and portable instruments	3	Understanding
M2.04	List the strategic planning and principles of energy management	3	Understanding
M2.05	Summarize the management of energy distribution and transmission losses	2	Understanding

M2.06	List the elements for energy monitoring and targeting	3	Understanding
M2.07	Identify the need for training program for energy management and energy management information systems.	2	Understanding
	Series Test – I		
Contents: Energy analysis – methodologies- block diagram- energy balance- laws of thermodynamics Energy management- principles of energy management- management of energy distribution- transmission losses. Energy measurement by portable and online instruments- electrical and temperature measurement meters Energy management information system – energy management training			
CO3	Identify the energy conservation plan and best financial investment.		
M3.01	Summarize the importance and principles of energy conservation	3	Understanding
M3.02	Summarize the implementation of planning for energy conservation	4	Understanding
M3.03	Summarize the energy conservation in paper, cement and sugar industry	4	Understanding
M3.04	Summarize the energy conservation for pumps, fans, boilers and air conditioning systems	4	Understanding
M3.05	Apply the basic principles in finding the best financial investment for energy conservation	3	Applying
Contents: Energy conservation –energy conservation vs energy efficiency-principles of energy conservation- Energy conservation plan- conservation of electrical energy, thermal energy- Energy conservation in paper industry, sugar industry and cement industry. Energy conservation for pumps, fans, boilers and air conditioning Energy economics – energy tariff-payback period-simple payback period method -net present value-internal rate of return- return on investment method-life cycle costing. [description with simple numerical]			
CO4	Summarize the energy auditing methods, Sankey diagram and total energy system.		
M4.01	Summarize the energy auditing methods.	3	Understanding
M4.02	Illustrate Sankey diagram for furnace, boilers	3	Understanding

M4.03	Summarize the instruments for energy auditing	3	Understanding
M4.04	Illustrate the total energy system with simple examples	2	Understanding
M4.05	Summarize the energy storage for process industries	1	Understanding
	Series Test – II		
Contents: Energy audit – types of audit-data for auditing- procedure for energy auditing- functions of energy auditor Sankey diagram- diagram for furnace and boiler Energy policy- total energy systems- examples Energy storage for process industries. – waste heat recovery- solar thermal generation-combined heat and power systems			

Text / Reference

T/R	Book Title/Author
T1	Energy Conservation and Management; S. S Thipse
R1	Non-conventional Energy Resources; S. Hasan Saeed
R2	Energy Management and Conservation; K V sharma, P. Venkateshaiah
R3	Energy engineering and management; AmlanChakrabarti

Online Resources

Sl.No	Website Link
1	https://nptel.ac.in/courses/112/105/112105221/